

1. Introduction and background

Because of climate change, the insurance industry has been facing increasingly unpredictable disaster claims, leading to higher operational costs. This challenge is particularly severe in the home insurance sector, contributing to the rising unaffordability of premiums. To offer customers coverage at a reasonable price while still maintaining a positive profit margin, actuaries can consider various approaches, or in this study-substituting premium payments with community service contributions or reducing policy coverage for less frequent disasters. This report aims to shed light on the first step of this process by qualitatively assessing the problem through modeling average premium payments based on area, thereby justifying their insurance affordability. The report starts by introducing its research background. Then, a series of qualitative steps will be taken to model average premium income, namely: Preliminary data analysis; Parametric curve fitting; and Graduation tests. Finally, a follow-up discussion about the chosen model will be provided.

The increase in natural disasters has impacted all regions of Australia, both financially and economically. These events affect not only residents and governments but also insurance companies. For individuals, the rise in claims due to catastrophic damage has led to higher premium costs, making home insurance unaffordable. The impact is especially severe in vulnerable households—who are often older, retired renters with lower savings and limited insurance literacy, as well as living in socioeconomically disadvantaged areas. According to the Australian Actuaries Home Insurance Affordability (AAHIA) Index, the most unaffordable premiums can cost 7.1 to 7.5 weeks of income, many times the median amount for those with lower pressures (1.0 week) (Actuarial Institute, 2024). This is partly because disadvantaged groups, concentrated in Northern Queensland, NT, and Northern NSW, face both catastrophic damage and low income. Additionally, the median AAHIA premium rises 0.4 to 10.7 weeks with each 3°C increase in global temperatures, and among households with premiums over \$2,000, half earn under \$65,000 (Actuarial Institute, 2024). To help tackle this problem, the report proposes 2 solutions, reducing premium through select scenario coverage and establishing social-work-for-premium program.

The two methods proposed aim primarily to reduce weekly premium payments, as well as utilizing word-of-mouth to spread insurance literacy.

Firstly, one viable solution is to limit scenario coverage, thus reducing the hazard rate, leading to the decrease in premium payment. Currently, most home insurance policies in Australia offer bundled coverage for storms, fires, and earthquakes, with flood cover as an add-on. While bundling may seem beneficial, as it offers broader coverage at a seemingly reduced rate, expanding the coverage scope increases the risk factor. Consequently, premiums for these bundles tend to be higher than premiums for principal coverage. Thus, by offering selective cover for most usual encountered disasters in a specific area, insurers can help their buyers save tons of money. One way of doing this is to offer a cost-plus pricing strategy, that is treating each risk premium as a specific amount and add up based on the policy. This offer is because one area is rarely exposed to more than 2-3 common risks, as can be seen in appendix A.1, A.2 and A.3.

Secondly, another proposal is to substitute premium cost with non-financial payment, or in this report- social work. This approach starts with creating a list of work that policy holders can take to reduce their periodic premium payments. The list of work may take the form as follows:

Type of work	Requirement	Premium reduction
Helping regional residents refurbish their houses	Need to attend 20 hours training and pass the training test	\$30 per hour per monthly premium payment
Using personal vehicles to help transport goods to demanded area	Have personal vehicles	\$32 per hour per monthly premium payment
Help with cooking and mental recovering for the disadvantaged	Need to attend 15 hours training and pass the training test	\$29 per hour per monthly premium payment

By contributing to the social program system, residents help ease the financial burden on local governments. The savings from reduced government spending could then be allocated to a common or government-owned reinsurance pool, which would help reduce insurers' claim burdens and, ultimately, drive down premium payments. According to the Department of Home Affairs Ministers (2024), federal disaster support provided to states has reached \$11.4 billion, funding prevention and recovery activities. Consider how much could be saved if more resources were contributed directly by residents themselves.

2. Preliminary data analysis

The first step is to change the column name of “Average.cost.of.4.week.premium” to “Training.Data” to match with the name format. Secondly, the dataset’s anomalies, namely invalid entries and negative values are addressed. Under this scenario, the author defines negative values as those average income lower than 0 and invalid data as those containing alphabetical characters that cannot be comprehended as a numeric value. In the report dataset, there are

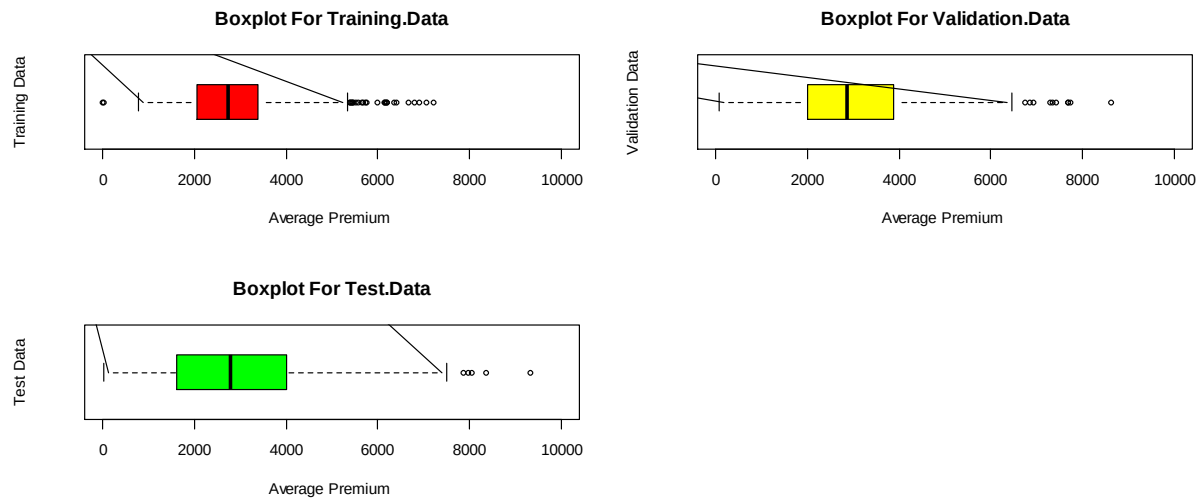
	Area.ID	Training.Data	Validation.Data	Test.Data
Invalid value	0	3	0	0
Negative Value	0	0	1	20

This report corrects negative income data by taking the absolute value and handles invalid or near-zero values by replacing them with the mean value of other valid entries in the same column. Next step, an exploratory data analysis of the dataset is performed:

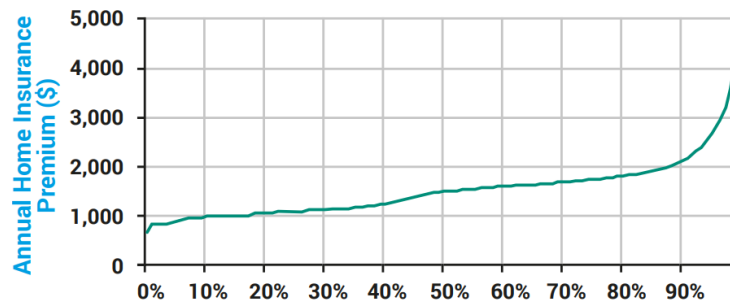
Area.ID	Training.Data	Validation.Data	Test.Data
Min. : 1.0	Min. : 3.39	Min. : 84.25	Min. : 27
1st Qu.:137.5	1st Qu.:2062.94	1st Qu.:1992.44	1st Qu.:1622
Median :274.0	Median :2744.25	Median :2876.38	Median :2780
Mean :274.0	Mean :2871.01	Mean :3007.62	Mean :2909
3rd Qu.:410.5	3rd Qu.:3390.67	3rd Qu.:3872.25	3rd Qu.:4022
Max. :547.0	Max. :7229.66	Max. :8621.66	Max. :9339

As can be seen from the data summary, our average premium dataset includes 547 areas sorted in increasing SEIFA score. The average premium per policy is split into 80-10-10 ration in training-validation-test data.

As the three datasets are all drawn from the same pool, a general summary can be drawn. At first sight, the distribution of average income tends to be right-skewed, which is explained by the presence of extreme outliers on the far right, significantly influencing the shape of the distribution. This suggests certain areas are disproportionately impacted by high premium costs, underscoring concerns about home insurance affordability. In terms of model building, while all three datasets have similar median values, the training data having a thinner interquartile demonstrates a more cluster around median value, indicating a predictability of parametric model fitting, though an outstanding number of extreme outliers may affect its accuracy. Conversely, the test and validation datasets, which contain fewer outliers and exhibit a broader median distribution, are more centered and stable. This indicates strong predictive performance and robustness in cross-validation. Boxplot depictions of the datasets are shown below:



In terms of the considerable number of outliers in training dataset, it is common practice to trim certain outliers. However, in this report, outliers are retained, as they represent an increased risk of premium unaffordability due to anticipated factors, such as rising disaster frequencies. According to the Actuarial Institute (2024), the AAHIA index shows a relatively stable premium payment in the first 9 quantiles while peaking in its final, showing that there is a huge premium payment difference among vulnerable regions and others, stressing risk of disaster exposure.



As the number of policy holders in a specific area is diverse and spread, it is possible to assume that the data given only capture a small fragment of its population for premium depiction. Thus, this phenomenon increases the volatility of estimates, bending the prediction line toward the observations rather than the true distribution (Home, 1972). By smoothing out these crude data using graduation, these effects could be somewhat mitigated, increasing model predictability.

3. Parametric model fitting

With an attempt to understand the impact of area disadvantages to premium cost, 3 approaches will be promoted to model Training.Data using Area.ID namely polynomial regression, natural cubic spline and smoothing spline. This conclusion is based on the distribution of the data, which can be reasonably approximated by fitting polynomial models on different intervals throughout the Area.ID (Appendix C.1)

Model 1: Polynomial regression

Using bare-eyed observations on the data, we can see that the training data seems to have 5 extremums, indicating that there might be a polynomial function with a degree of 4 for modelling. However, in order to terminate the effect of overfitting, the report will fit models with a degree from 1 to 6 and choose the one with the lowest MSE. Firstly, the simplest linear function will be modelled using generalized linear model with a gaussian distribution (as premium payment is numeric and can take different values). The R output indicates the model for training data has the form of:

$$\text{Training.Data} = 3991.475 - 4.089 \times \text{Area.ID}$$

This indicates a reverse correlation between the 2 variables, meaning that the more disadvantaged area (lower Area.ID) will suffer from higher premium payment. This model has an MSE of 880471.4, indicating its underfitting.

Secondly, the author will run a loop of data fitting for polynomial models having a degree of freedom from 2 to 8, the model may take the formula of:

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

With

- y: representing Training.Data
- x: representing Area.ID
- n: representing degree of the formula, which runs from 2 to 6

After executing the loop in R, we receive the following output:

```
with 1 increase in degree of freedom, MSE decreases by: 89754.28
polinimonal model with degree of freedom of 2 has an MSE of 790717.2

with 1 increase in degree of freedom, MSE decreases by: 32993.49
polinimonal model with degree of freedom of 3 has an MSE of 757723.7

with 1 increase in degree of freedom, MSE decreases by: 168819.1
polinimonal model with degree of freedom of 4 has an MSE of 588904.6

with 1 increase in degree of freedom, MSE decreases by: 52603.27
polinimonal model with degree of freedom of 5 has an MSE of 536301.3

with 1 increase in degree of freedom, MSE decreases by: 2324.381
polinimonal model with degree of freedom of 6 has an MSE of 533977

with 1 increase in degree of freedom, MSE decreases by: 27272.97
polinimonal model with degree of freedom of 7 has an MSE of 506704

with 1 increase in degree of freedom, MSE decreases by: 1396.604
polinimonal model with degree of freedom of 8 has an MSE of 505307.4
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According to the output, it seems that the higher degree of freedom correlates with better modelling, indicating that there might be many signals that polynomial models may fail to capture. However, a higher degree of freedom may also face the risk of overfitting. Thus, the author will choose the model with the highest MSE decline with 1 rise in model degree, which is the polynomial function with a degree of 4. Its formula is:

$$y = 2871 - 15102x + 7007x^2 - 4248x^3 + 9610x^4$$

Model 2: Natural Cubic Spline

Natural Cubic spline is defined as “a series of unique cubic polynomials are fitted between each of the data points” (McKinley & Levine, 1998). Thus, to form this model, the author will start with forming 8 intervals using 7 knots. There are 2 common systematic ways to choose knots: using Chebyshev spacing and based on the change of radius of curvature, this report will implement the former (Dung & Tjahjowidodo, 2017). In this report, a series of 7 nodes for Area.ID is formed using Chebyshev spacing (Appendix B), which are:

- X_1 : 7.84468
- X_2 : 60.56001
- X_3 : 155.54974
- X_4 : 274.00000
- X_5 : 392.45026
- X_6 : 487.43999
- X_7 : 540.15532

Next, the author will test if there is any optimal combination of these 7 knots to be used for generating natural cubic splines. This process is done through fitting each combination into the natural cubic spline model with an independent variable of Area.ID and a dependent variable of Training.Data. There are $2^7=128$ knot combinations (including an empty set). Then, Validation.Data will be used to compute each model’s MSE under the formula below:

$$MSE = \frac{1}{547} \sum_{t=1}^{547} (Validation.Data - Cubic Spline value)^2$$

and finally, the one with the lowest MSE will be chosen (details of the code can be found in the R script). The result is that the model with the lowest MSE (1279524) is the one with 6 nodes:

- X_1 : 7.84468
- X_2 : 60.56001
- X_3 : 155.54974
- X_4 : 274.00000
- X_5 : 487.43999
- X_6 : 540.15532

Model 3: Smoothing spline

Smoothing spline is defined as an estimate function drawn from a noisy dataset with an aim to balance the measure of goodness-of-fit (Wang, 2011). To form an optimal smoothing spline model, the author will start with estimating the optimal bias-variance tradeoff of the model. There are 2 ways to estimate this parameter-spar and lambda. However, in this study, the former will be used due to its nature of typical appearance in (0;1] interval, which help limit the testing range. To begin tuning this parameter, the report will set the number of spars tested, starting from 0 and ending at 1, with a jumping step of 0.01. Secondly, each spar will be used to create a smooth spline model using smooth.splines() function in R, with the predictor of Area.ID and the target of Training.Data. Next, the MSE of each model will be calculated using Validation.Data and the model with the lowest MSE will be chosen in the end. The result received is that the lower the spar value (closer to 0), the lower the MSE. However, as the spar value must be greater

than 0, the author will choose the model's spar at 0.01 (details will be found in the submitted R script).

Out-of-sample test

After having the 3 models, the author will compare their performances on Test.Data using the metric of MSE (the lower the more accurately predicted):

$$MSE \text{ each model} = \frac{1}{547} \sum_{t=1}^{547} (Test.Data - Predicted \text{ value})^2$$

With “Predicted value” as the fitted value of Area.ID in each of model. In the end, the author gets the following results: Model 1- 1903392; Model 2- 1791044 and Model 3- 1675495. In conclusion, the author decides that smooth spline model performs the best. The depictions of the 3 models can be found in appendix C.2. Moreover, during testing, the author has drawn out some key pros and cons for each model, as summarized below:

	Pros	Cons
Model 1	Have a smoothest line prediction (minimize overfitting) Capable of forming a formula for the model, which can be later used for traditional operations Can somewhat get key trends for certain intervals	Unsure if the chosen degree of freedom is the most optimal or not. There can be other formulas fitting the data that are not considered (log function, exponential function, Poisson distribution, etc.). Have the lowest performance on Test data of all 3 models
Model 2	Gradually smooth prediction line (low overfitting). Being able to capture the key extremums that reflect the trends for different intervals.	Extreme outliers may affect the assumptions of linearity on the far right of the prediction. Choice of knot is somewhat subjective and situational. There might exist higher degree (more than 3) that predict average premium in each interval
Model 3	Rough line prediction may indicate overfitting Capture almost every trend in any intervals	Spar value is set at a number close but not equals to 0, indicating that there is still a better spar (eg: 0.0001)

From the comparison, each model has its strength and weakness. However, as the report's primary concern is to provide regression on test dataset, model 3 is the most optimal option.

4. Graduation tests

Before going to each test, the z score, or the standardized deviation at each Area.ID is calculated:

$$z_c = \frac{\text{Training.Data} - \text{Predicted value}}{\sqrt{\text{Predicted value}}}$$

Where:

- Z_c is the z score for Area.ID = c
 - Training. Data and Predicted value are observed and calculated at Area.ID=c
- i. **Chi-squared test of fit**

H0: The smooth spline model aligns with actual Average premium payment for each area

H1: The graduated model does not align with actual Average premium payment for each area

The test score is calculated using the formula:

$$\chi^2 = \sum_{c=1}^{547} z_c^2 = 81886.66$$

This test score will be compared to the one-tailed critical value of 5%, at different degrees of freedom from 1 to 547. However, the score is larger than any calculated critical values, resulting in H0 rejection. This rejection can be explained using chart in appendix C.3, as the high test score is due to the high variance and lots outliers of the actual data.

ii. **Standardised deviations test**

H0: The number of observations in each interval follows a normal distribution.

H1: The number of observations in each interval does not follow a normal distribution.

The chosen intervals are $(-\infty; -3)$, $(-3; -2)$, $(-2; -1)$, $(-1; 0)$, $(0; 1)$, $(1; 2)$, $(2; 3)$, and $(3; +\infty)$. Then, the test the expected number of z score within each of the intervals $(-3; -2)$, $(-2; -1)$, $(-1; 0)$, $(0; 1)$, $(1; 2)$, and $(2; 3)$ is calculated as:

$$\text{number of expected } z_c = [P(X \leq \text{upper bound}) - P(X \leq \text{lower bound})] \times 547$$

The expected number of z scores within interval $(-\infty; -3)$ and $(3; +\infty)$ are equal and calculated as:

$$\text{number of expected } z_c = \frac{547 - [P(X \leq 3) - P(X \leq -3)] \times 547}{2}$$

The summary of these calculations is shown in the table below:

	$(-\infty; -3)$	$(-3; -2)$	$(-2; -1)$	$(-1; 0)$	$(0; 1)$	$(1; 2)$	$(2; 3)$	$(3; +\infty)$
Observation (O)	217	17	20	25	14	20	20	214
Expectation (E)	0.738	11.706	74.34	186.716	186.716	74.34	11.706	0.738

The test score can then be computed as:

$$\chi^2_7 \sim X = \sum_{c=1}^8 \frac{(O - E)^2}{E} = 125320.3 > 14.06714$$

The test score is larger than the critical value of 14.06714 (5% for degree of freedom of 7). Thus, the decision is to reject H0. This can be explained from the observations in the table above, where the data clusters at two ends rather than the center, contrasting to the normal distribution.

iii. Signs test

H0: The number of observations above the graduation line equals to that below the graduation line

H1: The number of observations above the graduation line does not equal to that below the graduation line

The sign test examines the number of positive and negative deviations of the actual data regarding the predicted model. The purpose of the test is to find the smallest value for k-the number of observations above the line- such that:

$$P(X \leq k) = \sum_{j=0}^k \binom{547}{j} \times 0.5^{547} \geq 0.025$$

Using R, the smallest k satisfies the above condition is 251. This calculation helps generate the upper critical value (547-251=296) and lower critical value (251) compared with the actual performance of z score (14+20+20+214=268). Thus, test fails to reject H0.

iv. Cumulative deviations test

H0: The overall number of observations matches with the overall number of predicted values

H1: The overall number of observations does not match with the overall number of predicted values

The test score is calculated as:

$$D_x = \frac{\sum_{j=1}^{547} (O - E)}{\sqrt{\sum_{j=1}^{547} E}} \sim N(0, 1)$$

The result calculated by R for this test is -1.857929e-13. Compared with the critical values at 5% rejection region for 2-tailed test, which is -1.96 and 1.96, the test score falls within the acceptance region. Thus, the test fails to reject H0.

v. Grouping of signs test

H0: The overall number of observed "clumping" of deviations matches with the overall number of predicted ones

H1: The overall number of observed "clumping" of deviations does not match with the overall number of predicted ones

Based on the observations from test 2, the write has computed the actual number of positive z score equals 268 and that of negative z score is 279. The purpose of the test is to find the smallest value for k-the number of groups above the line- such that:

$$P(X \leq k) = \sum_{t=1}^k \frac{\binom{268-1}{t-1} \binom{279+1}{t}}{\binom{547}{268}} = \sum_{t=1}^k \frac{\binom{267}{t-1} \binom{280}{t}}{\binom{547}{268}} \geq 0.05$$

The smallest k satisfies the above condition as calculated in R is 128. For the actual positive groups, the code idea is that for every sign change for z score from negative to positive when Area.ID increases by 1, the group sign will increase by 1 (can be found in the R script). Thus, the output received is G=160. As k=128<160, the test fails to reject H0.

vi. Serial correlations test-test at lag 1

The writer creates 2 samples from the z scores, the first one includes those with Area.ID from 1 to 546 (denoted $z^{(1)}$), the second one includes those with Area.ID from 2 to 547 (denoted $z^{(2)}$)

H0: the 2 samples are independent

H1: the 2 samples are not independent

The test score is:

$$r_1 = \frac{\sum_{i=1}^{546} (z_i - \bar{z}^{(1)})(z_i - \bar{z}^{(2)})}{\sqrt{\sum_{i=1}^{546} (z_i - \bar{z}^{(1)})^2 \sum_{i=1}^{546} (z_i - \bar{z}^{(2)})^2}} = -7.401833$$

Compared with the critical value at 5%, 2-tailed test, which is -1.64 to 1.64, the test rejects H0. The writer continues to test at different lag numbers and concludes that the smallest lag number to fail to reject H0 is 4 (details in the R script).

Even though the model retains its null hypothesis in 4 over 6 tests, the author determines that the smoothing spline is yet to be the best model. This is explained by the fact that the line is significantly rough, indicating an overfitting or undergraduation case. Even though when testing on Test.Data, the smoothing spline still results in a smaller MSE compared to the other 2 models, there is no concrete evidence that this performance can still be retained if the model is in another out-of-sample test. In short, this smoothing spline:

- lacks generalization ability
- lacks robustness across goodness-of-fit tests
- has limited flexibility with outliers and extremes
- has high variance in out-of-sample scenarios

5. Model discussion

a. Discuss the implications of the report findings for your approach/product.

The major implication of the report is that premium unaffordability or high premium payments do not only appear in disadvantaged areas but also occur in advantaged ones. This is evidenced by the existence of outliers in both extremely low and extremely high Area IDs. When introducing a product that substitutes premium payments with social work, it is crucial for insurance companies and the government to recalculate the Socio-Economic Indexes for Areas (SEIFA) scores to avoid the misallocation of resources (Area ID has nearly the same definition as SEIFA in this case).

Additionally, it is concerning that areas with high SEIFA scores still experience peak premium payments despite being expected to have a favorable living environment. This situation proves that disaster prevention work has not yet achieved effectiveness. Thus, governments must relocate their funding toward prevention rather than recovery activities to minimize the damage of increasing disasters.

For insurance providers, the studies help determine the possible areas of social work that policyholders can engage in. One implication of this approach is that the severity of disaster damage may vary among different areas, even though the severity of premium unaffordability is the same. We can assume that the presence of high-cost premiums correlates with a higher claims ratio, thus more disaster damage. Consequently, by locating help and social work in demanded areas, insurers can help with distributing human resources to the right place at the right time, which in turn not only reduces the cost of government spending but also impacts insurance literacy. Policyholders joining charity work may spread their goals to their relatives and those who are helped in disaster-devastated areas, thus utilizing word-of-mouth to spread information.

Another implication is that different areas may have different severities of disaster. Insurers should specify each premium deduction bracket for each region, thus making policyholders' work fairer and more incentivized.

b. Discuss any potential improvements that could be made to the models.

General Improvement:

One projected improvement is to add one or more additional variables to the model of premium payment regression. This is because the purpose of the report is to understand the effect of high premium payments and their distributions. Therefore, this study focuses on outliers. In fact, one variable is not enough to capture the distribution of these extreme values. By identifying and incorporating other factors besides Area ID, the model's accuracy and predictive power could be enhanced (Pearson, 2002; Kalisch et al., 2016). In other words, by broadening the range of variables, the model can provide a more comprehensive understanding of the underlying phenomena, leading to more reliable predictions.

Secondly, locating Area ID ranges with similar trends based on socioeconomic knowledge is highly recommended. This might involve redefining boundaries or categories within the Area ID to reflect more nuanced regional differences (Brauner & Shacham, 1998). Such adjustments can help identify specific patterns and anomalies that broader categorizations might obscure, leading to a more precise understanding of premium unaffordability factors in both advantaged and

disadvantaged areas. Moreover, these interval trends are more intuitive in terms of common knowledge, thus increasing the model's interpretability.

Thirdly, the current six tests for outlier detection focus primarily on central distribution. My suggestion is to use other tests besides the goodness-of-fit, such as those based on robust statistics or machine learning techniques, which could provide a more comprehensive identification of outliers (Swedish Defence Research Agency, 2019). These advanced methods can detect outliers that traditional tests might miss, thereby improving the model's robustness and reliability. Thus, by incorporating these techniques, the model becomes more adaptive to handle diverse datasets and improves overall performance.

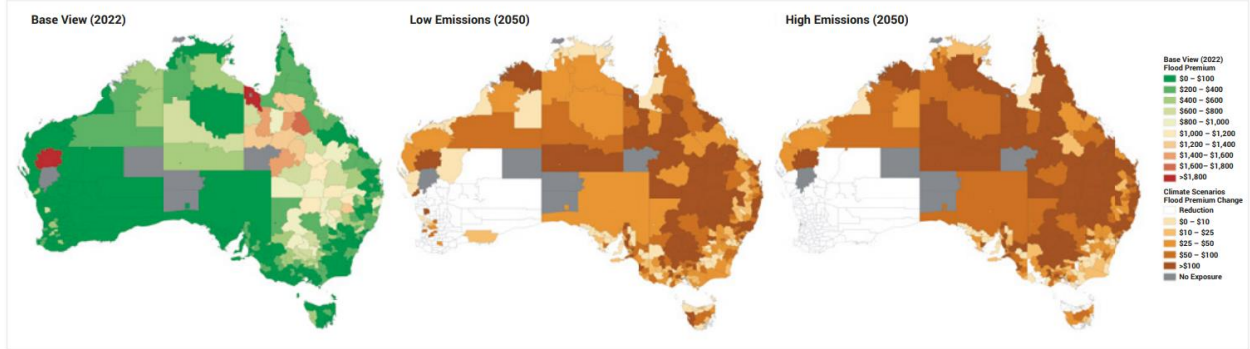
Fourthly, there is a slight chance that there might not exist a parametric model that explains the correlation between Area ID and Average Income. Therefore, a suggestion for future studies is to fit nonparametric models like KNN to test if the accuracy of the prediction is improved.

Lastly, implementing metrics to assess the robustness of the models can aid in selecting the one with the most optimal variance-bias trade-off. This involves evaluating the model's performance under different conditions to ensure it generalizes well to new data (Xu & Yang, 2024).

Robustness metrics can provide insights into the model's stability and reliability, helping to choose models that maintain high performance across various scenarios. By focusing on these areas, the models can be refined to provide more accurate and reliable predictions, ultimately leading to better decision-making and resource allocation.

6. Appendix

Appendix A.1: Average flood premiums by LGA – current, and changes under low and high emissions scenarios (\$2022 values) (Actuarial Institute, 2024)



- n is the total number of nodes, or in this case is 7
- x_k is the k^{th} node in the $(-1;1)$ interval. Thus, k can be any integer value from 1 to 7

Adopting the above formula, a series of 7 nodes in $(-1;1)$ is formed, which are:

- x_7 : 9.749279e-01
- x_4 : 6.123032e-17
- x_1 : -9.749279e-01
- x_6 : 7.818315e-01
- x_3 : -4.338837e-01
- x_5 : 4.338837e-01
- x_2 : -7.818315e-01

Following on, the node value will be scaled to the Area.ID size using affine transformation:

$$X_k = \frac{\max(\text{Area.ID}) - \min(\text{Area.ID})}{2} \times x_k + \frac{\max(\text{Area.ID}) + \min(\text{Area.ID})}{2}$$

- X_k is the k^{th} knot of the natural cubic spline
- $\text{Max}(\text{Area.ID})$ is the highest value of Area.ID, which is 547
- $\text{Min}(\text{Area.ID})$ is the lowest value of Area.ID, which is 1
- x_k is the k^{th} node in the $(-1;1)$ interval

Thus: $X_k = 273 \times x_k + 274$

Adopting the above formula, a series of 7 nodes for Area.ID is formed, which are:

- X_1 : 7.84468
- X_4 : 274.00000
- X_7 : 540.15532
- X_2 : 60.56001
- X_5 : 392.45026
- X_3 : 155.54974
- X_6 : 487.43999

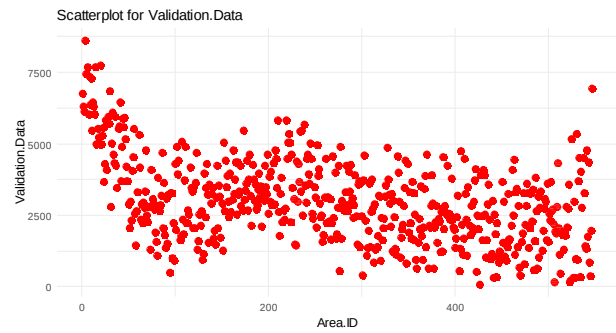
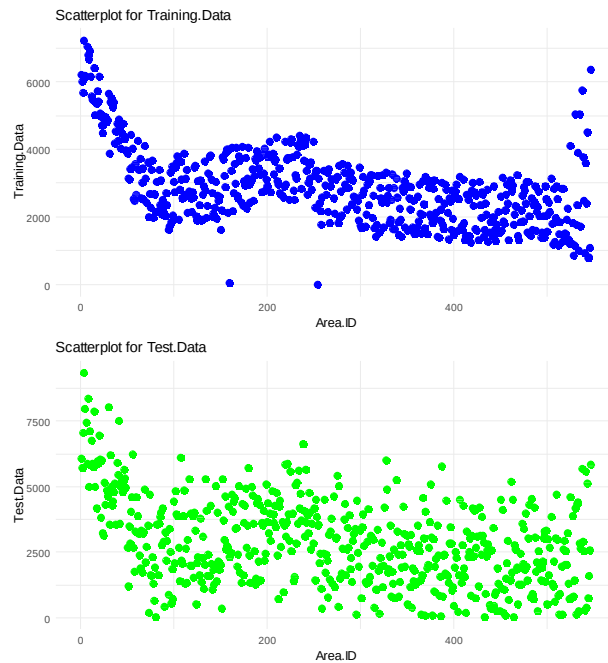
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$$MSE = \frac{1}{547} \sum_{t=1}^{547} (\text{Tra.Data} - \text{Predict value})^2$$

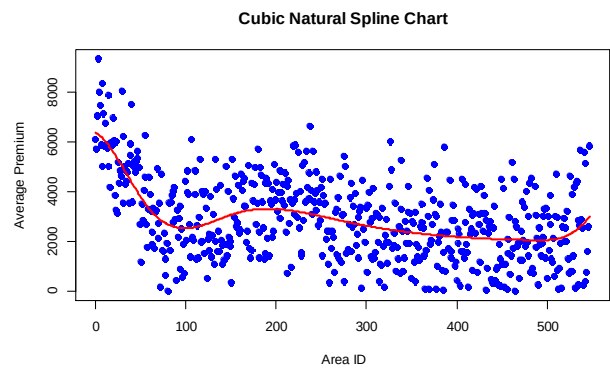
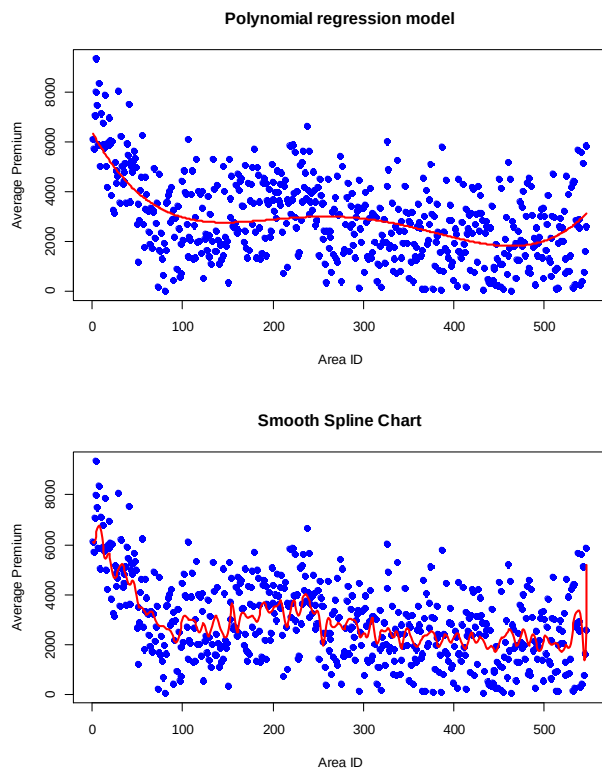
and finally, the one with the lowest MSE will be chosen (details of the code can be found in the R script). The result is that the model with the lowest MSE (1279524) is the one with 6 nodes:

- X_1 : 7.84468
- X_3 : 155.54974
- X_5 : 487.43999
- X_2 : 60.56001
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- X_6 : 540.15532

Appendix C.1: Scatter plot for the distribution for Training, Validation and Test Data

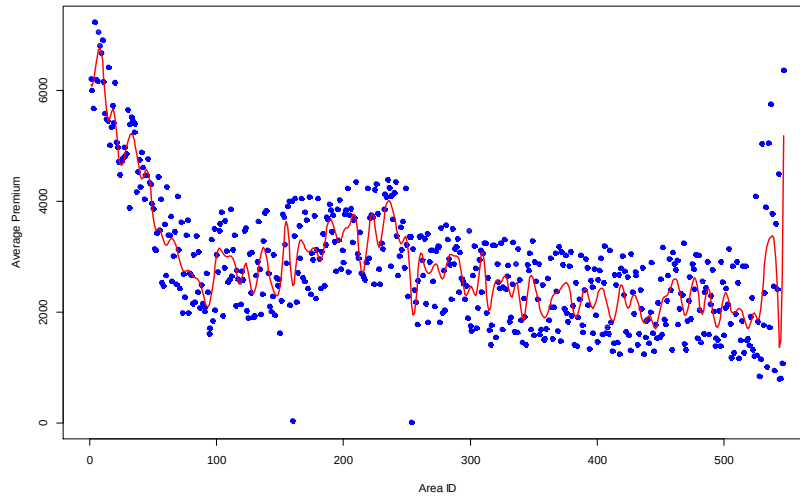


Appendix C.2: Performance of each model on Test.Data



Appendix C.3: Performance of smoothing model on Training.Data

Smooth Spline Chart



7. Reference

- Actuaries Institute (2024). *Home insurance affordability and socioeconomic equity in a changing climate*. [online] Default. Available at: <https://www.actuaries.asn.au/public-policy-and-media/our-thought-leadership/reports/home-insurance-affordability-and-socioeconomic-equity-in-a-changing-climate>.
- Australian Bureau of Statistics. (2021). Socio-Economic Indexes for Areas (SEIFA), Australia. Retrieved from <https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/latest-release1>.
- Australian Bureau of Statistics. (2021). Using and interpreting SEIFA. Retrieved from <https://www.abs.gov.au/statistics/detailed-methodology-information/concepts-sources-methods/socio-economic-indexes-areas-seifa-technical-paper/2021/using-and-interpreting-seifa>
- Dung, V. T., & Tjahjowidodo, T. (2017). A direct method to solve optimal knots of B-spline curves: An application for non-uniform B-spline curves fitting. *PloS one*, 12(3), e0173857.
- Hoem, J. M. (1972, January). On the statistical theory of analytic graduation. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability (Univ. California, Berkeley, Calif., 1970/1971), Theory of statistics Volume I* (pp. 569-600).
- McKinley, S., & Levine, M. (1998). Cubic spline interpolation. *College of the Redwoods*, 45(1), 1049-1060.
- Swedish Defence Research Agency. (2019). *Evaluation of robustness metrics*. Swedish Defence Research Agency. https://www.foi.se/download/18.7884c07518a3b5f5b5d86c/1700157898644/Evaluation_of_Robustness_Metrics_FOI-S--6692--SE.pdf
- Taylor, P. (2024). *How much does home insurance cost in Australia?* [online] finder.com.au. Available at: <https://www.finder.com.au/home-insurance/home-insurance-cost>.
- Wang, Y. (2011). *Smoothing splines: methods and applications*. CRC press.
- Xu, X., & Yang, Y. (2024). A critical review of automated machine learning applications in bioinformatics. *BioTech*, 8(1), Article 8. <https://doi.org/10.3390/biotech8010008>
- Pearson, R. K. (2002). Outliers in process modeling and identification. *IEEE Transactions on control systems technology*, 10(1), 55-63.
- Kalisch, M., Michalak, M., Sikora, M., Wróbel, Ł., & Przyszałka, P. (2016). Influence of outliers introduction on predictive models quality. In *Beyond Databases, Architectures and Structures. Advanced Technologies for Data Mining and Knowledge Discovery: 12th International Conference, BDAS 2016, Ustroń, Poland, May 31-June 3, 2016, Proceedings II* (pp. 79-93). Springer International Publishing.
- Brauner, N., & Shacham, M. (1998). Role of range and precision of the independent variable in regression of data. *AIChE journal*, 44(3), 603-611.

